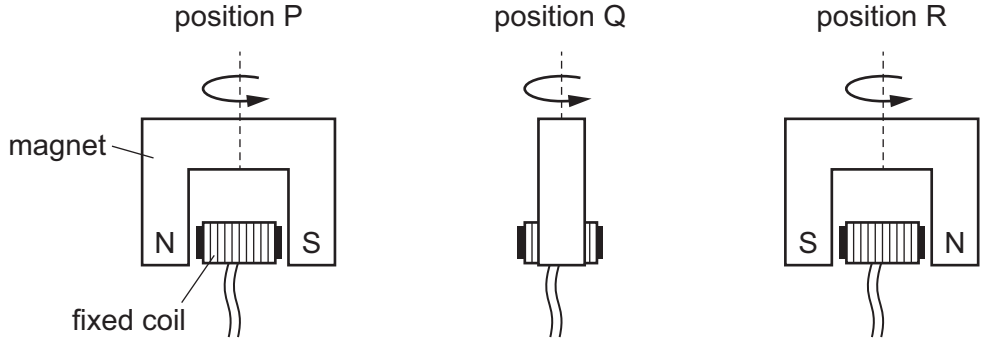
31 The diagram shows a simple a.c. generator. The magnet rotates around a fixed coil wound round a fixed iron core. The electromotive force (e.m.f.) output is a sine wave, a series of crests, zeros and troughs.

The magnet is shown in three positions, P, Q and R, during its rotation. The magnetic field in the coil is zero at Q but changing most rapidly.



Which statement about the induced e.m.f. in the coil is correct?

A The magnitude of the e.m.f. at position Q is larger than that at position P.

B The magnitude of the e.m.f. at position Q is zero.

C The magnitude of the e.m.f. at position Q is smaller than at position R, but not zero.

D The magnitude of the e.m.f. at position R is larger than at position P.